NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*COSC(MODEL-A&APEX)	RTA-4	Date: 01-03-26
Time: 3 Hrs	2025_P2	Max. Marks: 180

KEY SHEET

MATHEMATICS

1	**D**	2	**B**	3	**A**	4	**C**	5	**AC**
6	**CD**	7	**AC**	8	**ABCD**	9	**49**	10	**36**
11	**23.50**	12	**3.20**	13	**125**	14	**4.50**	15	**2**
16	**3**								

PHYSICS

17	**B**		**D**	19	**B**	20	**B**	21	**AD**
22	**BCD**	23	**BD**	24	**ABC**	25	**25**	26	**8**
27	**2**	28	**4.90**	29	**10**	30	**0.25**	31	**2.24**
32	**4**								

CHEMISTRY

33	**D**	34	**C**	35	**A**	36	**C**	37	**ACD**
38	**ABD**	39	**ABD**	40	**ABD**	41	**72**	42	**2**
43	**126**	44	**8**	45	**28**	46	**12**	47	**4.54**
48	**5**								

Rankers Academy JEE

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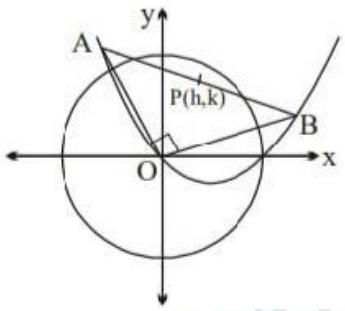
SOLUTIONS

MATHS

1. Equation of chord of the circle with mid-point (h, k) is

$hx + ky = h^2 + k^2$

On homogenizing the parabola, we get



$$x^2 - \frac{(x+y)(hx+ky)}{h^2+k^2} = 0$$

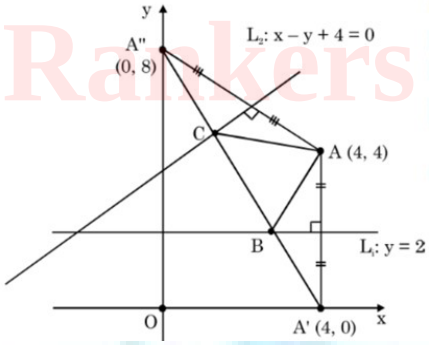
$$\Rightarrow (h^2+k^2)x^2 - (x+y)(hx+ky) = 0$$

Now coefficient of x^2 + coefficient of $y^2 = 0$

$$\therefore \quad h^2 + k^2 - h - k = 0$$

$$\Rightarrow \quad x^2 + y^2 - x - y = 0 \Rightarrow \text{circle with center } \left(\frac{1}{2}, \frac{1}{2}\right) \text{ and radius } = \sqrt{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\sqrt{2}}$$

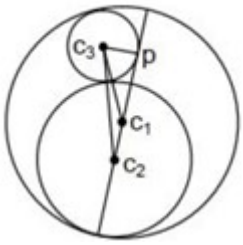
2.



Equation of $A'A''$ is $2x + y = 8$

$$B \equiv (3,2) C \equiv \left(\frac{4}{3}, \frac{16}{3}\right)$$

3.



$$c_1 : \left(\frac{-5}{\sqrt{2}}, \frac{-5}{\sqrt{2}}\right)$$

$r_1 = 5$

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$c_2 : \left(\frac{-3}{\sqrt{2}}, \frac{-3}{\sqrt{2}}\right)$

$r_2 = 3$

r be radius of third circle and its center be at c_3, then

$$(c_2c_3)^2 = (pc_3)^2 + \left(c_1c_2 + \sqrt{\left((c_1c_3)^2 - (pc_3)^2\right)}\right)^2$$

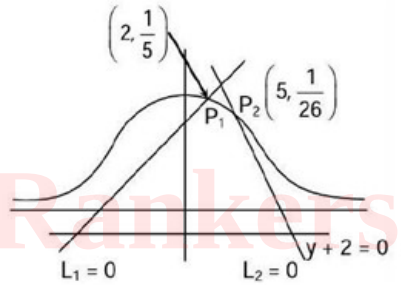
$$\Rightarrow (r_2 + r)^2 = r^2 + \left(r_1 - r_2 + \sqrt{(r_1 - r)^2 - r^2}\right)^2$$

$$\Rightarrow r = \frac{4r_1r_2(r_1 - r_2)}{(r_1 + r_2)^2} = \frac{15}{8}$$

4. As $P\left(\alpha, (1+\alpha^2)^{-1}\right)$ lie on $y = \frac{1}{1+x^2}$

$\therefore$ On solving $y = \frac{1}{x^2+1}$ with L_1, we get $P_1\left(2, \frac{1}{5}\right)$

And with line L_2, we get $P_2\left(5, \frac{1}{26}\right)$



$\therefore\ 2 \le \alpha \le 5$

5. S_2 is the circle with given points as diameter, so. S_1 is $x^2 + y^2 - 2x - 2y = 0$, which passes through origin. Clearly S_2 is $x^2 + y^2 + 2x + 2y = 0$.

Lies it be (h,-h) then $\sqrt{(h-1)^2 + (-h-1)^2} = 2\sqrt{2} \Rightarrow h = \pm\sqrt{3}$

Centre of S_3 is $\left(\sqrt{3}, -\sqrt{3}\right)$ or $\left(-\sqrt{3}, \sqrt{3}\right)$.

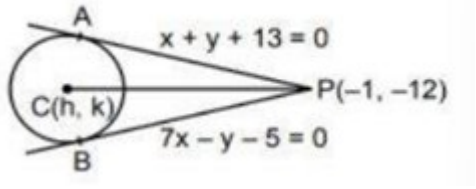
One of the equations of S_3 is $x^2 + y^2 - 2\sqrt{3}x + 2\sqrt{3}y + 4 = 0$

Common tangent of S_1 and S_2 is $x + y = 0$

And that of S_1 and S_3 is $\left(\sqrt{3}-1\right)x - \left(\sqrt{3}+1\right)y - 2 = 0$ .So the radial centre is $\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$, say P

then radius of the circle intersecting S_1, S_2 and S_3 orthogonally $OP = \sqrt{\frac{2}{3}}$

6.



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Clearly, the point lies on $7x - y = 5$

Also, center of the circle must lie on the bisectors of the lines

$x + y + 13 = 0$ and $7x - y - 5 = 0$ given by

$$\frac{x+y+13}{\sqrt{2}} = \pm\frac{7x-y-5}{\sqrt{50}} \Rightarrow x - 3y = 35 \text{ or } 3x + y + 15 = 0$$

Let (h, k) be the center of the circle, then h-3k=35…….(1)

And 3h+k=15…..(2)

Clearly CB is perpendicular BP $\Rightarrow \frac{K-2}{h-1} \times 7 = -1 \Rightarrow h + 7k - 15 = 0$ $\quad \ldots..(3)$

On solving, we get centers as $C_1 = (29-2)$ and $C_2 = (-6,3)$

$\Rightarrow r_1^2 = 800$ and $r_1^2 = 50 \Rightarrow$ smaller circle has radius $= \sqrt{50}$

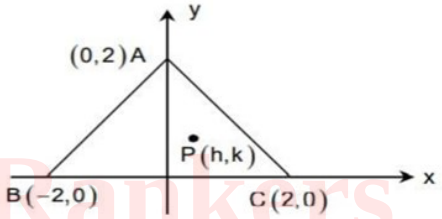
Area of quadrilateral ACBP=$2\left[\frac{1}{2} \times \sqrt{50} \times \sqrt{200}\right]$ sq units

7. $AB : y - x = 2$

$AC : x + y = 2$

$\therefore$ For P (h, k)

$$\left|\frac{h+k-2}{\sqrt{2}}\right|\left|\frac{h-k+2}{\sqrt{2}}\right| = 2k^2$$



$$\frac{x^2}{16/3} + \frac{(y+2/3)^2}{(16/9)} = 1 \qquad \therefore e = \sqrt{\frac{2}{3}}$$

8. Equation of tangent at P is x+y=4

Clearly mirror image of R(-5,5) lies on line PQ.

Now mirror image R' of R

$$\Rightarrow \frac{\alpha' + 5}{1} = \frac{\beta' - 5}{1} = \frac{-2(-5+5-4)}{2} = 4$$

(α', β')=(-1,9)

Let PM cuts the axis at T; As M is midpoint of PT,T is (-3, 7)

We know that SP = ST and $\angle SMP = \frac{\pi}{2}$

Equation SP is y-3 = $-\frac{1}{3}(x-1)$

$= x + 3y - 10 = 0$

Let $S(10-3\beta, \beta)$

Again TS || PQ $\Rightarrow \frac{\beta - 7}{13 - 3\beta} = \frac{9-3}{-1-1} = -3$

$\therefore$ focus is (-2, 4)

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9. $\Delta_1 =$ Area of APQR

$$= \frac{1}{2}\begin{vmatrix} a\cos\theta_1 & b\sin\theta_1 & 1 \\ a\cos\theta_2 & b\sin\theta_2 & 1 \\ a\cos\theta_3 & b\sin\theta_3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}ab\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix} \text{ and}$$

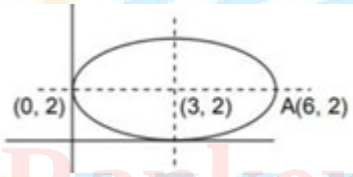
$\Delta_2 =$ Area of Δ P`Q'R

$$= \frac{1}{2}\begin{vmatrix} a\cos\theta_1 & a\sin\theta_1 & 1 \\ a\cos\theta_2 & a\sin\theta_2 & 1 \\ a\cos\theta_3 & a\sin\theta_3 & 1 \end{vmatrix} = \frac{1}{2}a^2\begin{vmatrix} \cos\theta_1 & \sin\theta_1 & 1 \\ \cos\theta_2 & \sin\theta_2 & 1 \\ \cos\theta_3 & \sin\theta_3 & 1 \end{vmatrix}$$

Clearly, $\frac{\Delta_1}{\Delta_2} = \frac{1}{7} \Rightarrow \frac{\Delta_2}{\Delta_1} = 7$

10. $C_1 : \frac{(x-3)^2}{9} + \frac{(y-2)^2}{4} = 1$

$\therefore A_1 = (6,2), A_2 = (0,2)$



$\therefore |PA_1 - PA_2| = 3\sqrt{2}$

Clearly locus of P is hyperbola for which

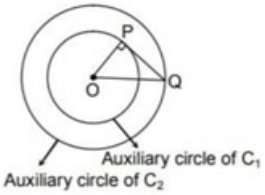
$A_1A_2 = 2ae = 6$ and $2a = 3\sqrt{2} \Rightarrow e = \sqrt{2}$

$\therefore$ Locus of P is rectangular hyperbola

Equation of conic $C_2 : (x-3)^2 - (y-2)^2 = \frac{9}{2}$

Now $D_1 = 2ae = 6$

$D_2 = b^2 = \frac{9}{2}$



$D_3 = \sqrt{9 - \frac{9}{2}} = \frac{3}{\sqrt{2}}$

$\therefore \left(\frac{D_1D_2}{D_3^2}\right)^2 = 36$

11. Locus is a hyperbola with eccentricity $= = \frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$

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12. Since ellipse touches x-axis at O(0,0), y-axis is therefore normal to be ellipse.

$\Rightarrow$ Y-axis is the internal angular bisector of $\angle F_1OF_2$ (F1 and F2 being the two foci)

$\Rightarrow$ Equation of OF_1 and OF_2 can therefore be assumed as y=kx and y=-kx

Equation of major axis is $y = x + 1$. Solving OF_1 and OF_2 with major axis.

We get $F_1 = \left(\frac{1}{k-1}, \frac{k}{k-1}\right)$ and $F_2 = \left(\frac{-1}{k+1}, \frac{k}{k+1}\right)$

C= midpoint of $F_1F_2 = (3,4)$

$= K = \frac{2}{\sqrt{3}} \Rightarrow F_1 = \left(3 + 2\sqrt{3}.4 + 2\sqrt{3}\right)$ and $F_2 = \left(3 - 2\sqrt{3}.4 - 2\sqrt{3}\right)$

$CF_1 = ae = \sqrt{\left(2\sqrt{3}\right)^2 + \left(2\sqrt{3}\right)^2} = 2\sqrt{6}$

Also major axis

$= 2a = OF_1 + OF_2 = \sqrt{\left(3 + 2\sqrt{3}\right)^2 + \left(4 + 2\sqrt{3}\right)^2} + \sqrt{\left(3 - 2\sqrt{3}\right)^2 + \left(4 - 2\sqrt{3}\right)^2} = 4\sqrt{7}$

$= a = 2\sqrt{7} \Rightarrow e = \sqrt{\frac{6}{7}}$

13. $A, B = \left(1 \pm \sqrt{10}\left(-\frac{1}{\sqrt{10}}\right), 5 \pm \sqrt{10}\left(\frac{3}{\sqrt{10}}\right)\right)$

$= (0,8), (2,2)$

$\Delta CAB = 4 \Rightarrow ab = 4$

If θ is angle between asymptotes, then $\cos\theta = \frac{3}{5} \Rightarrow \cos\frac{\theta}{2} = \frac{2}{\sqrt{5}} \Rightarrow e = \frac{\sqrt{5}}{2}$

14. $P = \left(at_1^2, 2at_1\right), Q = \left(-at_1^2, 0\right), R = \left(at_2^2, 2at_2\right)$ where $t_2 = -t_1 - \frac{2}{t_1}$

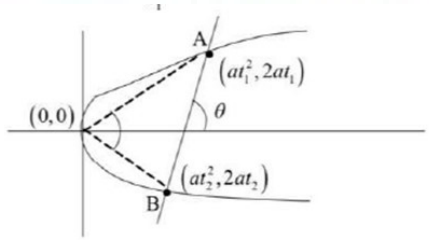
$P\left(2a, 2\sqrt{2}a\right), Q(-2a, 0), R\left(8a, -4\sqrt{2}a\right)$

$PQ = 2\sqrt{6}a$, $PR = 6\sqrt{3}a$, $QR = 2\sqrt{33}a$, Area of $\Delta PQR = 18\sqrt{2}a^2$

15. $y^2 = 8x$

Now, $t_1t_2 = -4$

And $t_2 = -t_1 - \frac{2}{t_1}$



$\frac{-4}{t_1} = -t_1 - \frac{2}{t_1} \Rightarrow -t_1^2 = -2$

$\Rightarrow t_1^2 = -2$

$\Rightarrow t_1^2 = 2$

$y^2 = 4ax$

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$$\frac{dy}{dx}=\frac{4ax}{2y}$$

$$\frac{dy}{dx}=\frac{1}{t}$$

Slope of normal $=-t$

And $\tan^2\theta=t_1^2=2$

16. Given that $\frac{2b^2}{a}=2\Rightarrow\frac{2a^2}{a}=2\Rightarrow a=1$

Let $S^1(x,y)$ be the other focus

$\Rightarrow|SO-S^1O|$

$\Rightarrow\sqrt{x^2+y^2}-1=\pm2$

$\Rightarrow\sqrt{x^2+y^2}-1=\pm2$

$\Rightarrow\sqrt{x^2+y^2}=3$

PHYSICS

17. After the switch 'S' is shifted to position-2,

$$L_1\frac{di}{dt}+L_2\frac{di_2}{dt}=0$$

$$Ldi_1+4Ldi_2=0$$

$$di_1+4di_2=0$$

$$\int_{t_2}^{1}di_1+4di_2=0$$

$i_1+i_0=-4i_2\Rightarrow i_1+4i_2=i_0$ .....(i)

Where the charge on the capacitor is maximum, (i)=0

Hence $i_1=i_2$

From (i) and (ii), we get

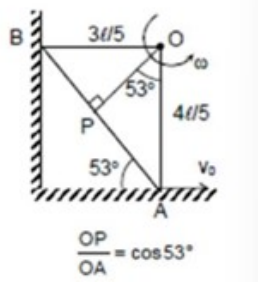
$$i_1=i_2=\frac{i_0}{5}=\frac{\varepsilon}{5R}\qquad\left(i_0=\frac{\varepsilon}{R}\right)$$

Now, $\frac{q^2{}_{max}}{2C}=\frac{1}{2}Li_0^2-\frac{1}{2}(L+4L)\left(\frac{i_0}{5}\right)^2$

$$\frac{q^2{}_{max}}{2C}=\frac{2Li_0^2}{5}\Rightarrow q_{max}=\frac{\varepsilon}{R}\sqrt{\frac{4CL}{5}}$$

18. V_A-V_P is maximum when OP is perpendicular to AB (O is IAOR)

$$\omega=-\frac{v_0}{4l/5}=\frac{5v_0}{4l};\ |v_A-v_P|_{max}=\frac{B\omega}{2}\left[(OA)^2-(OP)^2\right]$$



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$$= \frac{B_0}{2}\frac{5v_0}{4l}\left[\left(\frac{4l}{5}\right)^2 - \left(\frac{12l}{25}\right)^2\right]; \text{OP} = \frac{3}{5}\left(\frac{4l}{5}\right) = \frac{12l}{25}$$

$$= \frac{5B_0v_0l^2(20+12)(20-12)}{8l(25)^2} = \frac{32}{125}B_0lv_0$$

19. Let R_S is resistance of the loop filed through loop $B = (\mu_0 R\sigma\omega)/2 = (\mu_0\sigma R\alpha t)/2$

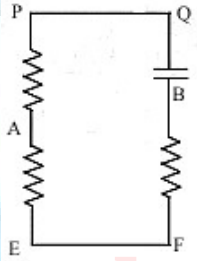
Induced emf in the loop $e = (\pi r^2\mu_0\sigma R\alpha)/2$

Current in the loop $i = \frac{e}{R_s} = \frac{\pi r^2\mu_0\sigma R\alpha}{2R_S}$

$$F = iBr = \frac{\pi r^2\mu_0\sigma R\alpha}{4R_s}\mu_0\sigma R\alpha tr$$

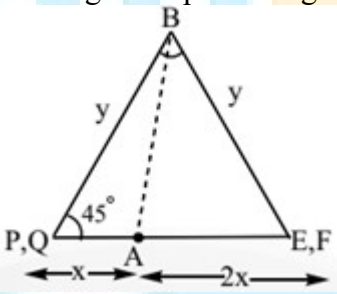
$$F = \left(\frac{\pi\mu_0^2\sigma^2\alpha^2R^2r^3}{4R_S}\right)t$$

20. Let voltage phasors along be PA, AE be x, 2x



And voltage phasors along QB, BF be y,y (as $R = \frac{1}{\omega C}$)

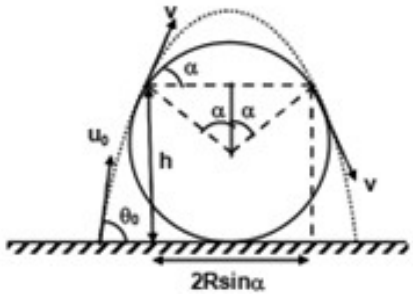
The phasors diagram representing them can be drawn as shown



So phasor length AB (i.e; reading of V_2) is $= \sqrt{x^2 + y^2 - 2xy\cos 45^0}$ using cosine rule in ΔPAB)

Also $x + 2x = V$ and $\sqrt{2}y = V \Rightarrow$ Reading of $V_2 = \frac{V}{3}\sqrt{\frac{5}{2}}$

21. For u to be minimum, it just grazes the cylinder at two points as shown.
From COME



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$$\frac{mu^2}{2}=\frac{mv^2}{2}+mgh$$

$$u^2=v^2+2gh$$

$$u^2=v^2+2gR(1+\cos\alpha)$$

Also $2R\sin\alpha=v^2\frac{\sin 2\alpha}{g}\Rightarrow v^2=\frac{Rg}{\cos\alpha}$ .....(2)

From (1) and (2)

$$u^2=Rg\left[2+2\cos\alpha+\frac{1}{\cos\alpha}\right]$$

For $u\to$min. $\frac{d}{d\alpha}(u^2)=0\Rightarrow\cos\alpha=\frac{1}{\sqrt{2}}\Rightarrow\alpha=45^0$

$$u_0^2=Rg\left[2+2\sqrt{2}\right]$$

Also $u\cos\theta=v\cos\alpha$

$$\tan^2\theta_0=\left[3+2\sqrt{2}\right]$$

22. Suppose that at any instant of time, velocity of the rod is v toward right
The current in the circuit is i. In the figure,

$V_a-V_b-V_d-V_c$

i.e Ldi=Bldx

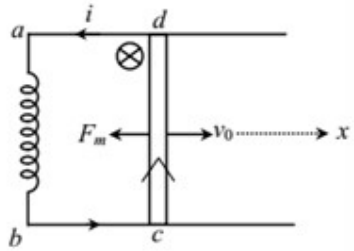
Integrating, we get Li=Blx

Magnetic force on the rod at this instant is

$$F_m=i\ell B=\frac{B^2\ell^2}{L}x \quad .....(i)$$

Since, this force is in opposite direction of $\vec{v}$, so from Newton`s second law can write,

$$m\left(\frac{d^2x}{dt^2}\right)=-\frac{B^2\ell^2}{L}x$$



Comparing this with equation of SHM, i.e

$$\frac{d^2x}{dt^2}+\omega^2x=0\Rightarrow x=A\sin\omega t\Rightarrow v=A\omega\cos\omega t....(ii)$$

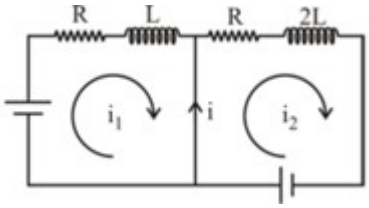
We, have $\omega=\frac{Bl}{\sqrt{mL}}$

So $T=\frac{2\pi}{\omega}=2\pi\frac{\sqrt{mL}}{Bl}=2\pi\frac{\sqrt{8\times2}}{\pi\times2}=4$ s

At t=0, $v=v_0\Rightarrow v_0=A\omega\Rightarrow A=\frac{v_0}{\omega}=\frac{4}{\pi}$m

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23.



$i_{max} = (i_2 - i_1)_{max}$

$\Delta i = (i_2 - i_1)$

$= \frac{V}{R}\left[1 - e^{-\left(\frac{R}{2\ell}\right)t}\right] - \frac{V}{R}\left[1 - e^{\left(-\frac{R}{L}\right)t}\right]$

For $(\Delta i)_{max}$ $\frac{d(\Delta i)}{dt} = 0$

$\frac{V}{R}\left[-\frac{R}{L}e^{-\left(\frac{R}{L}\right)T} - \left(-\frac{R}{2L}\right)e^{-\left(\frac{R}{2L}\right)t}\right] = 0$

$e^{-\left(\frac{R}{2L}\right)t} = \frac{1}{2}e^{-\left(\frac{R}{2L}\right)t} \Rightarrow e^{-\left(\frac{R}{2L}\right)t} = \frac{1}{2}$

$\left(\frac{R}{2L}\right)t = \ell n2$

$t = \frac{2L}{R}\ell n2 \rightarrow$ time when i is maximum

$i_{max} = \frac{V}{R}\left[e^{-\frac{R}{L}\left(\frac{2L}{R}\ell n2\right)} - e^{-\left(\frac{R}{2L}\right)\left(\frac{2l}{R}\ell n2\right)}\right] \Rightarrow \quad |i_{max}| = \frac{V}{R}\left|\left[\frac{1}{4} - \frac{1}{2}\right]\right| = \frac{1}{4}\frac{V}{R}$

24. $|z_1| = \sqrt{x_c^2 + R^2} = 2R, z_1 = R + \sqrt{3}Rj \quad |z_2| = \sqrt{(x_C - x_L)^2 + (2R)^2} = 2R, z_2 = 2R$

$z_p = \frac{z_1 z_2}{z_1 + z_2} = \frac{2R^2(1+\sqrt{3})j}{\sqrt{3}R(\sqrt{3}+1)j} = \frac{2R}{\sqrt{3}}$, wattles Current in given by Component of RMS Current

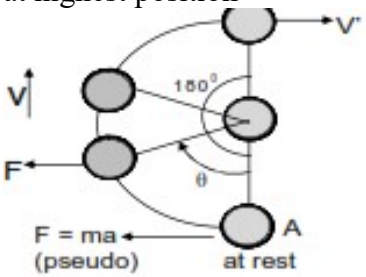
perpendicular to Voltage pate Phasor $I_{wattless} = \frac{V_0}{\sqrt{2}(2R)}\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}V_0}{4\sqrt{2}R}$

25. Acceleration of train $= a \rightarrow$

Motion of bob w.r.t. train:

Initially $\vec{V}_{bob/train} = O$

From $A \rightarrow B$ pseudo force F=ma $\leftarrow$ acts on bob. Let bob turns through 180^0 and has speed V` at highest position



W/E theorem $A \rightarrow B$

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$$FR=\frac{1}{2}mv^2+mgh$$

$$\frac{mv^2}{R}=2(ma-mg) \quad .....(i)$$

$B\rightarrow C$: Now train has stopped, a=0

No pseudo force acts on bob

Conservation of Energy

$$mgR+\frac{1}{2}mv'^2=\frac{1}{2}mv^2$$

$$\frac{mv'^2}{R}=\frac{1}{2}mv'^2=\frac{1}{2}mv^2$$

$B\rightarrow C$: Now train has stopped, a=0

No pseudo force acts on bob.

Conservation of Energy

$$mgR+\frac{1}{2}mv'^2=\frac{1}{2}mv^2$$

$$\frac{mv'^2}{R}=\frac{mv^2}{R}-2mg$$

T+mg=2 (ma-mg)-2mg [from eq. (i)]

On solving $a\geq 25m/s^2$

26. $N=mg\cos\theta+m\omega^2R=mg\cos\theta+m\left(\frac{5g}{R}\right)R\Rightarrow N=5mg+mg\cos\theta>0$, so block does not leave circular motion

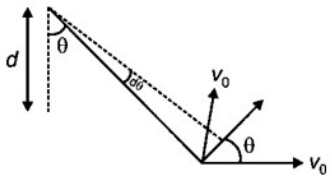
$f_r=ma_T$ For limiting case $\mu N=ma_T$

$$\Rightarrow \quad \mu\left(mg(5+\cos\theta)\right)=mg\sin\theta\Rightarrow\mu=\frac{\sin\theta}{5+\cos\theta}$$

$$\frac{d\mu}{d\theta}\frac{(5+\cos\theta)(\cos\theta)-\sin\theta(-\sin\theta)}{(5+\cos\theta)^2}=0\Rightarrow\cos\theta=-\frac{1}{5}$$

$$\mu_{max}=\frac{\sqrt{1-\frac{1}{25}}}{5-\frac{1}{5}}=\frac{2\sqrt{6}}{24}\Rightarrow x=8$$

27.



$$dt=\frac{2d\,d\theta}{2v_0\cos\theta}$$

$$t=\frac{d}{v_0}\int_0^{\pi/3}\sec\theta d\theta$$

$$=\frac{d}{v_0}\ln\left|(\sec\theta+\tan\theta)\right|_0^{\pi/3}$$

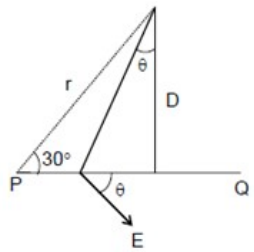
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$$= \frac{d}{v_0}\ln\left(\sqrt{3}+2\right) - \ln(1)$$

$$= \frac{d}{v_0}\ln\left(2+\sqrt{3}\right)$$

28. $E2\pi r = \pi r^2 \frac{dB}{dt}$

$$\Rightarrow E = \frac{B_0 r}{2}$$



$$a_r = \frac{qE}{m}\cos\theta = \frac{qB_0 D}{2m}.$$

$$3\ell = \frac{1}{2}\frac{qB_0 D}{2m}t^2$$

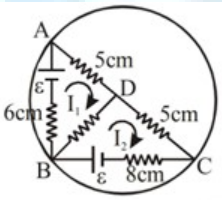
$$t = \sqrt{\frac{12\ell}{\left(\frac{qB_0}{m}\right)D}} \qquad .....(i)$$

$$\frac{D}{(t/2)} = \tan 30^0 \Rightarrow D = \frac{\ell}{2\sqrt{3}} = 1$$

$$t = \sqrt{24} = 4.90 \text{ s}$$

$$v^2 = 2a_\tau(3\ell) = 2\times\frac{qB_0 D}{2m}\times 3\times 2\sqrt{3} = 4.24 \text{ m/s}$$

29. Induced EMF in the loop ABD or BDC $= \frac{d\phi}{dt} = A\frac{dB}{dt}$



$$\varepsilon = \frac{1}{4}\times 8\times 6\times 10^{-4}\times 0.263 = 12\times 0.263\times 10^{-4}$$

Say the resistance per cm of wire is R

Now applying KVL to loops ADB and BDC.

$$5I_1 + 5(I_1 - I_2) + 6I_1 = \frac{\varepsilon}{R} \text{ or } 16I_1 - 5I_2 = \frac{\varepsilon}{R}$$

Similarly $13I_2 - 5(I_1 - I_2) = \frac{\varepsilon}{R}$ or $18I_2 - 5I_1 = \frac{\varepsilon}{R}$

Solving this we get, $i_1 = \frac{23}{263}\frac{\varepsilon}{R}$ or $18I_2 - 5I_1 = \frac{\varepsilon}{R}$

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$$\Rightarrow i_1 - i_2 = \frac{2}{263}\frac{\varepsilon}{R} = \frac{2}{263}\times 12\times 0.263\times 10^{-4}\times\frac{1\times 1\times 10^{-6}}{24\times 10^{-9}\times 1\times 10^{-2}} = 0.01A = 10mA$$

30. The torque applied by the magnetic force F_m about O is given $d\tau_m = r\left(dF_m\right) = r\left\{(dr)B\right\}$

$$= \tau_m = \int d\tau_m = \int r(idrB) : \text{Putting } i = \frac{E}{R} = \frac{B\ell^2\omega}{2R}$$

We obtain, $\tau_m = \left(\frac{B^2t^2\omega}{2R}\right)\int_0^{\ell} rdr\ \tau_m = \frac{B^2\ell^2\omega}{2R}\frac{r^2}{2}\Big|_0^1 \quad \Rightarrow \tau_m = \frac{B^2\ell^4\omega}{4R}$

The external torque must be equal and opposite to the magnetic torque $\Rightarrow \tau_m = \frac{B^2\ell^4\omega}{4R}$

31. Induced emf in loop BENM

$$\varepsilon_1 = A_2\frac{dB}{dt} = 1\times 2 = 2V$$

Induced emf in loop CFNM

$$\varepsilon_2 = A_2\frac{dB}{dt} = 0.5\times 2 = 1V$$

Induced emf in loop ABCD

$$\varepsilon_3 = A_3\frac{dB}{dt} = 1.5\times 2 = 3V$$

Applying loop rule in ADFE

$$3 - 7l_1 - 7l_2 + 3 = 0$$

$$\Rightarrow l_1 + l_2 = \frac{6}{7} \quad .....(i)$$

Applying loop rule in ADCB

$$3 - 7l_1 - 3(i_1 - i_2) = 0$$

$$\Rightarrow 10l_1 - 3l_2 = 3 \quad .....(ii)$$

After solving equation (i) and (ii)

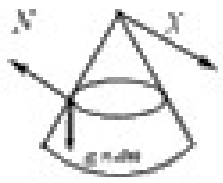
$$l_1 = l_2 = \frac{3}{7}$$

Applying loop rule in MNEF

$$2 - 4l_2 + (l_1 - l_2)\times 2 - \frac{q}{C} = 0$$

$$\Rightarrow \frac{q}{C} = 2 - 4\times\frac{3}{7} = \frac{2}{7} \Rightarrow q = \frac{2}{7}\times 7.84 = 2.24\ \mu C$$

32.



Consider the potential energy of the band as a function of R. To be in equilibrium, the band must be at a position of minimum potential energy. Let the elastic potential energy be

$U_1 = \frac{1}{2}k\left(2\pi R - 2\pi R\right)^2$ and the gravitational potential energy measured with respect to the vertex of the code $U_1 = -mg(R - r)\cot\theta$

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The total potential energy is $U = 2k\pi^2(R-r)^2 - mg(R-r)\cot\theta$

At the minimum position,

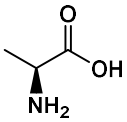
$$\frac{dU}{dR} = 4\pi^2 k(R-r) - mg\cot\theta = 0$$

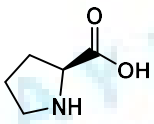
Which has solution

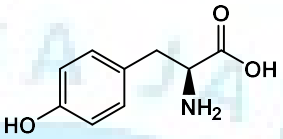
$$R = r + \frac{mg}{4\pi^2 k}\cot\theta$$

CHEMISTRY

33.

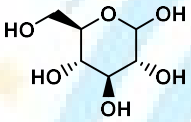
(i) 

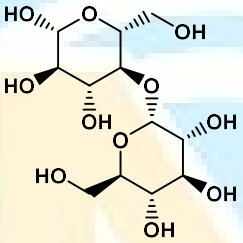
(iii) 

(v) 

are non-essential amino acids

34.

(i) 

(v) 

give Fehling's due to hemiacetal linkage

35. 1 → B: Froth flotation is the standard physical method to concentrate sulfide ores (uses collectors/frothers).
2 → A: Roasting oxidizes ZnS to ZnO with release of SO_2 ($2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$).
3 → C: Calcination means heating in absence/limited supply of oxygen (used for carbonates/oxides); it is not the standard step for ZnS → Zn (roasting is used instead).
4 → D: Carbon reduction (pyrometallurgy) reduces ZnO to metallic Zn (zinc vapour collected by condensation).
5 → E: Electrolytic refining produces high-purity zinc by electrolysis of purified zinc solutions (e.g., $ZnSO_4$).

36. Chromyl chloride formation (balanced):

1. $4\,NaCl + K_2Cr_2O_7 + 6\,H_2SO_4 \rightarrow 2\,CrO_2Cl_2 + 2\,KHSO_4 + 4\,NaHSO_4 + 3\,H_2O$

(A) =CrO_2Cl_2 → atoms per molecule = Cr(1) + O(2) + Cl(2) = 5.

2. Hydrolysis of chromyl chloride:

$$CrO_2Cl_2 + 4\,NaOH \rightarrow Na_2CrO_4 + 2\,NaCl + 2\,H_2O$$

(B) =Na_2CrO_4 → atoms per molecule = $Na_{(2)}$ + Cr(1) + $O_{(4)}$ = 7.

3. Formation (and isolation) of the peroxo species under cold ether conditions:

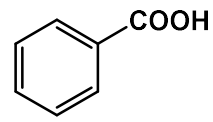
$$Na_2CrO_4 + H_2SO_4 + 2\,H_2O_2 \xrightarrow{0-5\ ^\circ C,\ \text{ether}} CrO_5 + Na_2SO_4 + 3\,H_2O$$

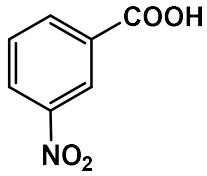
(C) =CrO_5 → atoms per molecule = Cr(1) + O(5) = 6.

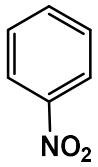
Total atoms = $5 + 7 + 6 = 18$.

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37.

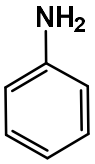
X = 

Y = 

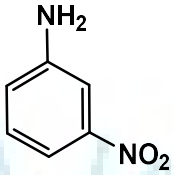
Z = 

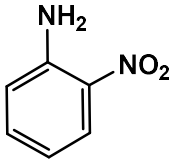
Compound 'Z' does not undergo Friedel-craft reaction

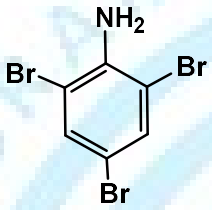
38.

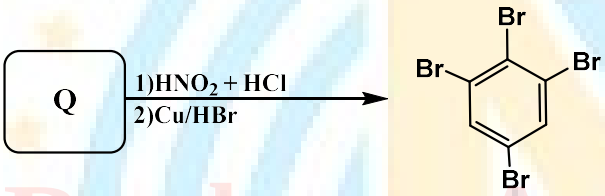
W = 

X = P = 

Y = 

Z = 

Q = 

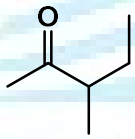


(major)

39. The rule is: a reducing couple (CO/CO_2) can reduce any oxide whose formation line lies **above** the CO/CO_2 line at that temperature. From the data: CO/CO_2 intersects AO at 900 K and BO at 1300 K, so at 1000 K the CO/CO_2 line lies between AO and BO (i.e., below BO). Therefore CO is thermodynamically able to reduce BO at 1000 K.
CO/CO2 line has a positive slope

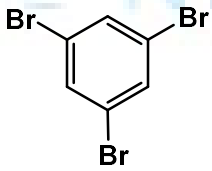
40. $E^0_{Cr^{3+}/Cr^{2+}} < E^0_{Fe^{3+}/Fe^{2+}} < E^0_{Co^{3+}/Co^{2+}}$

41.

P = 

42. It contains three acetyl groups and two propanoyl groups.

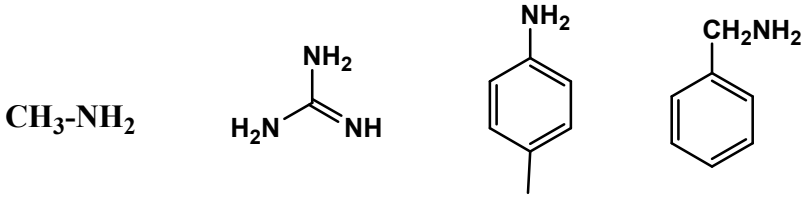
43.

E = 

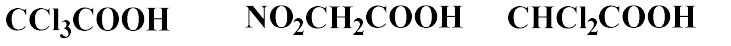
Weight of product E = 0.4 x 315 = 126

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44.



are more acidic than aniline



are more acidic than formic acid

45. Dissolves in aerated KCN → forms $[Ag(CN)_2]^-$
refined by cupellation → **Silver (Ag)**
M_2: Forms $M_2(CO)_4$ at 330–350 K → **Nickel (Ni)**
(Mond process: $Ni(CO)_4$)
M_3: Forms volatile tetraiodide in Van Arkel → **Titanium (Ti)** (TiI_4)
A=2 ; B=1 ; C=3 ; D=22

46. The oxide is MnO_2.
Oxidation states:
$Mn: +4 \rightarrow +2$
$\rightarrow$ gains 2 electrons
So, X = 2
Chloride oxidation: $2C\ell^- \rightarrow Cl_2 + 2e^-$
Thus, **2 moles of HCl** act as reducing agent (their Cl^- gets oxidized
$Y = 2 \Rightarrow N = 5(2) + 2 = 12$

47. Alkaline: $+7 \rightarrow +6$, so $\Delta = 1$.
Thermal: $2KMnO_4 \rightarrow O_2$, so $G = 1$.
Electron gain: acidic $= 5$, neutral $= 3$, so $R = 5/3$.
$M_{KMnO_4} = 158$, hence 15.8 g = 0.1 mol → $MnO_2 = 0.1 \times 87 = 8.7$g →
$M/10 = 0.87$.
Thus $N = 1 + 1 + 5/3 + 0.87 = 4.54$

48. **Concentration:** Froth flotation (since it is a sulphide ore).
Roasting:

$$2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2$$

Reduction:

$$ZnO + C \xrightarrow{\Delta} Zn + CO$$

Refining:
– *Electrolytic refining* (Zn anode, pure Zn cathode, acidified $ZnSO_4$).
– *Distillation* (low b.p. 907°C allows separation from Cd, Pb).

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